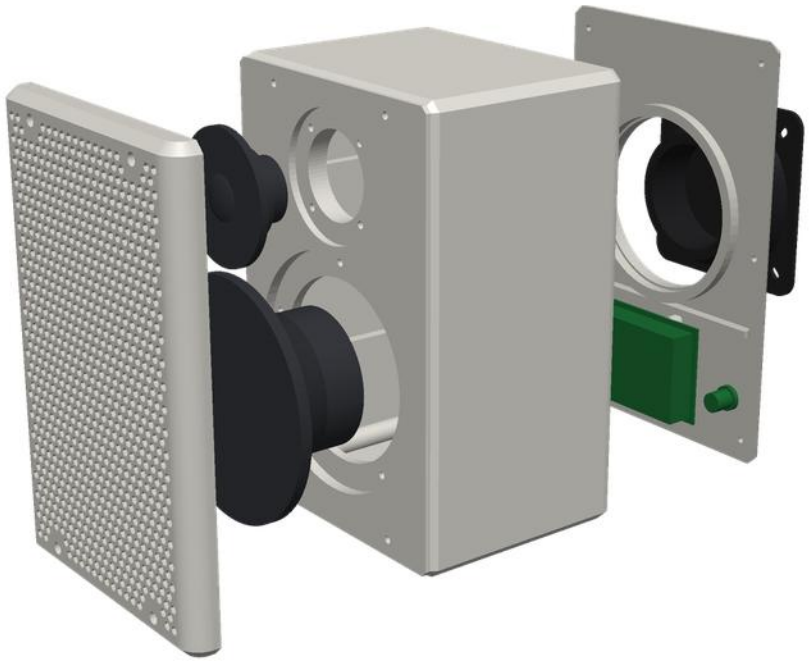
ESP-ONE v5 – Picture Guide



Parts per speaker

- **26×** Insert M3×5
- **6×** M3×10
- **9×** M3×12
- **8×** M3×8
- **3×** M3×6

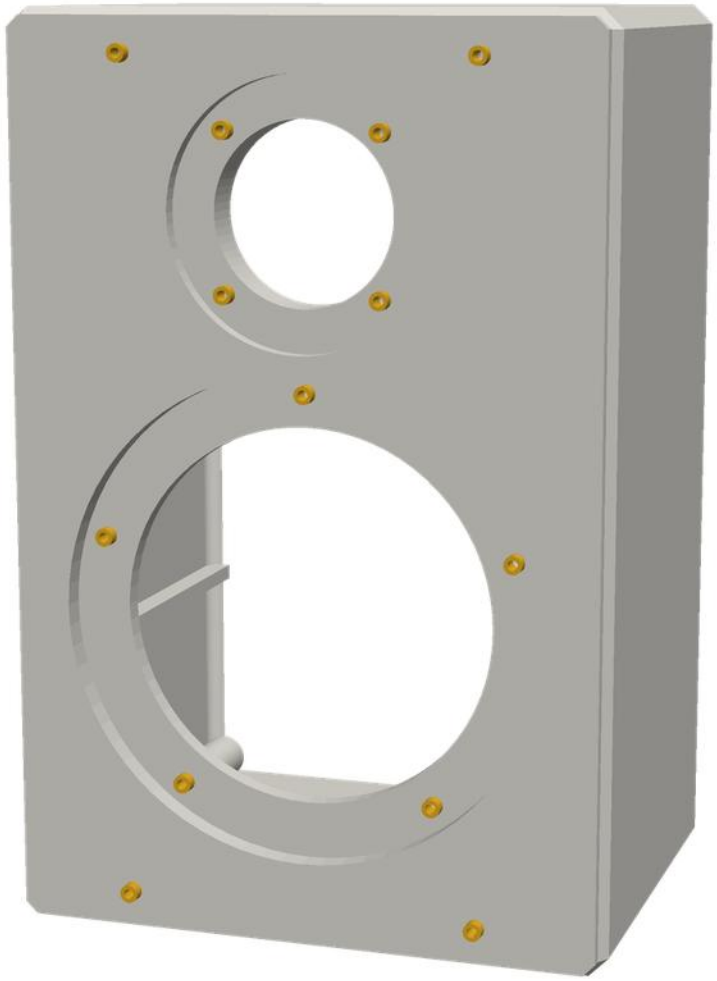
Heat-set insert: M3 × L5 × OD4.5
Screws: socket head (2.5 mm hex key)

Printed parts: body, back panel,
front grille (+ test-fit piece)

Drivers, PCB, power supply,
gasket tape 1 + 2 mm, damping

Tools: soldering iron · 2.5 mm hex key · small screwdriver · multimeter · calipers · kitchen scale

1 Heat-set inserts – body front



5× Woofer

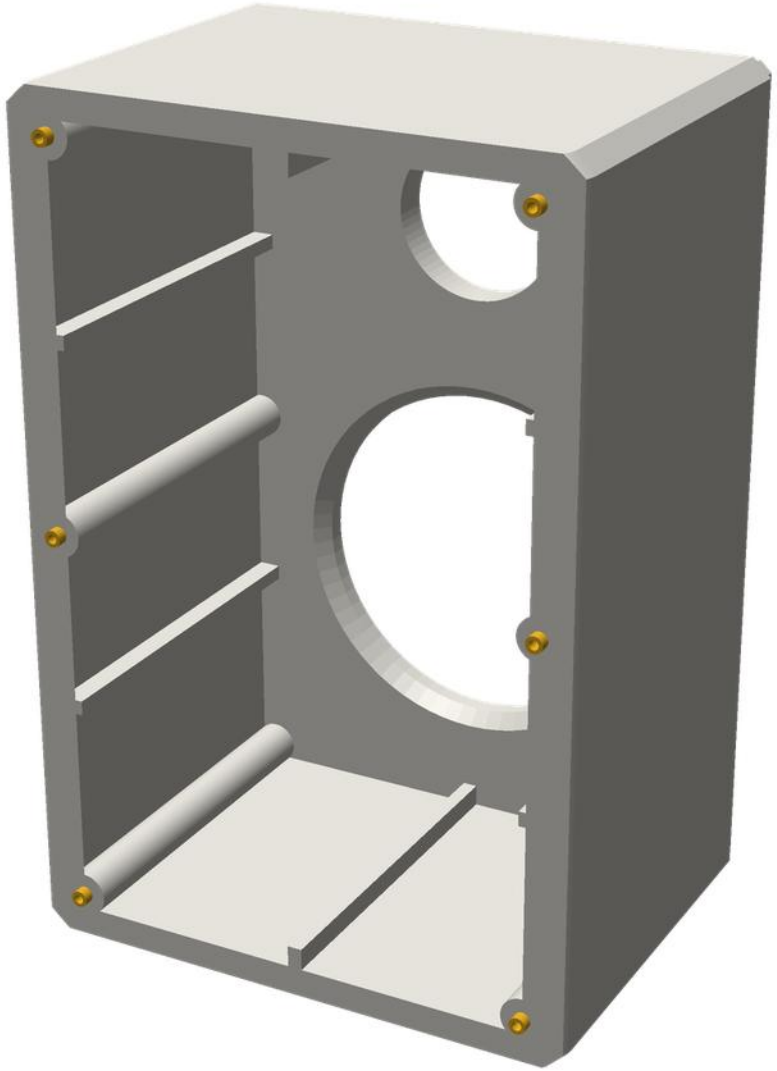
4× Tweeter

4× Front grille

= 13 inserts

Soldering iron 220–230 °C (PLA 200 °C)
Place the insert straight,
let it sink in without force,
finish flush.

2 Heat-set inserts – body back



6× Pillar ends

4 corners + 2 on the sides
(for the back panel)

3 Heat-set inserts – back panel

4× Passive radiator



outside

3× PCB



inside

4 Fit the back panel

3× M3×6 PCB

PCB: components facing inwards

Radiator from the outside,
1 mm gasket under the flange

4× M3×8 radiator



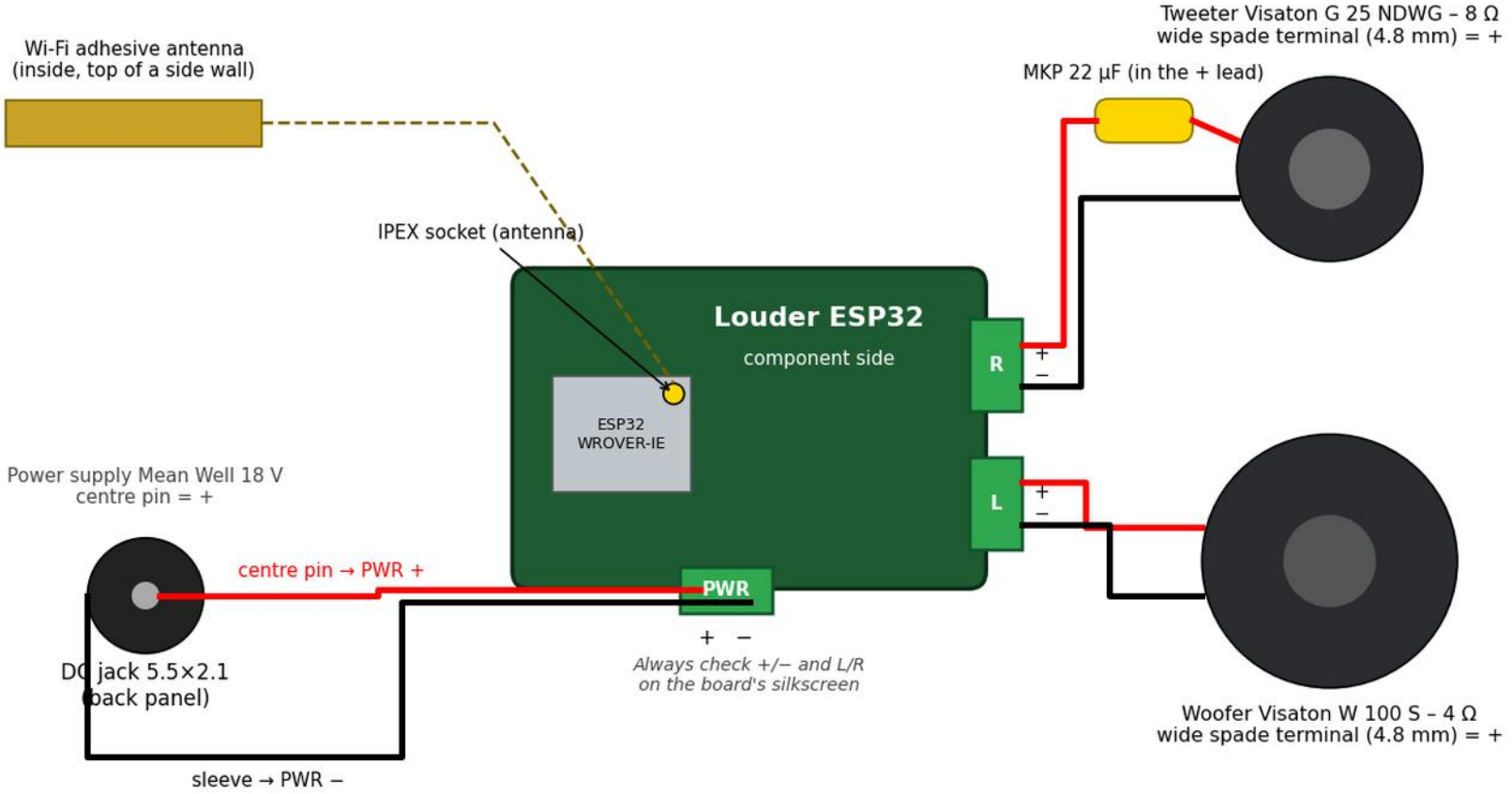


First fit the DC jack (nut inside) · stick butyl mat inside (not on PR ring / standoffs) · plug antenna into IPEX

5 Wiring

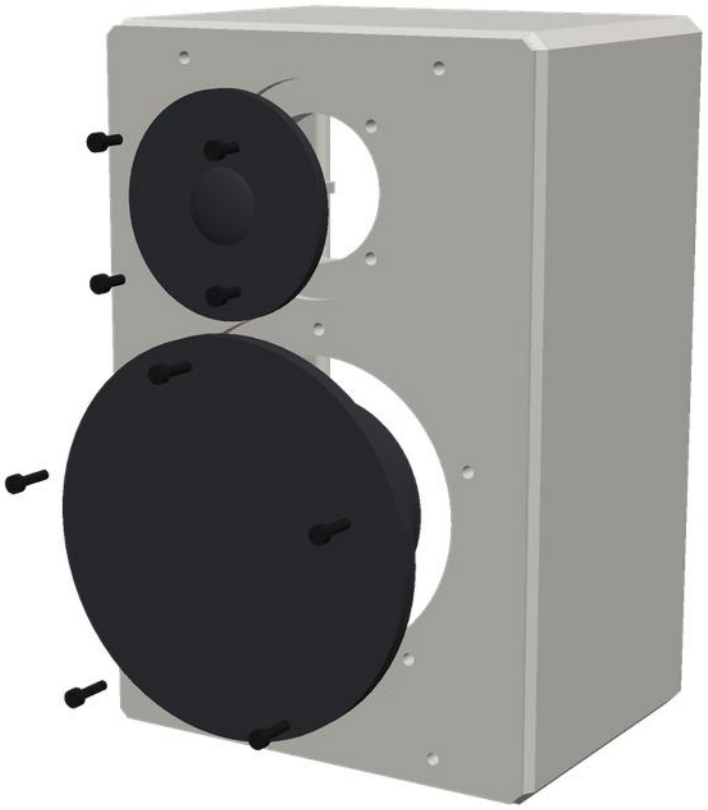
Wiring ESP-ONE – always + to +

Wires: red = +, black = − · Lengths: woofer approx. 20 cm, tweeter approx. 30 cm (along the side wall), DC approx. 10 cm
Solder the wires to the drivers (or use 4.8 / 2.8 mm spade connectors), insulate joints and capacitor with heat shrink



! Only connect the tweeter once the ESP-ONE DSP settings are loaded.
Check the power supply polarity with a multimeter first: centre pin = +18 V.

6 Fit the drivers



5× M3×12 woofer

4× M3×8 tweeter

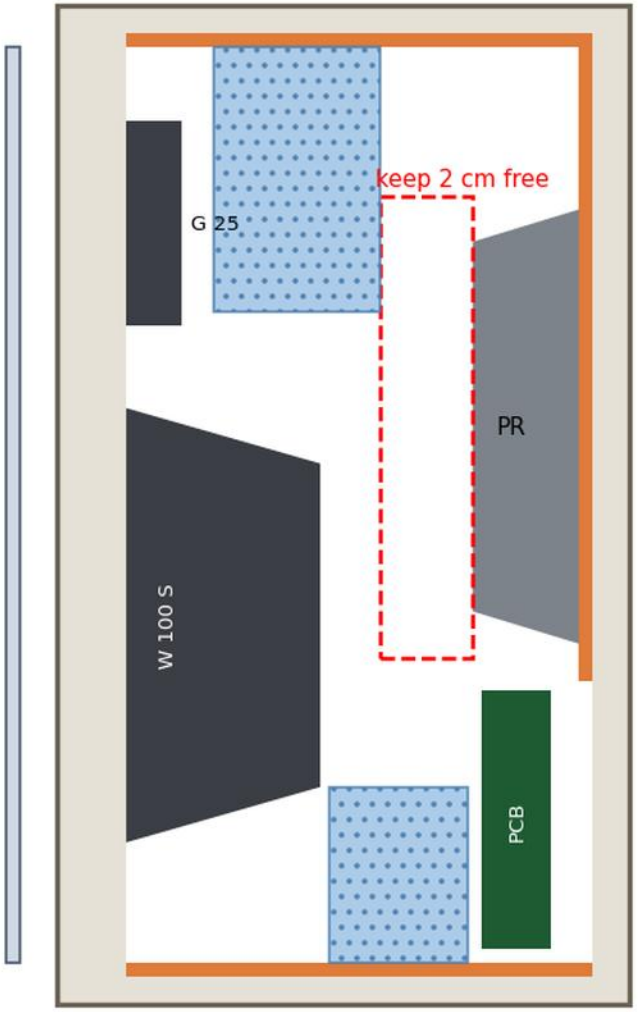
Before:

- Solder the wires to the drivers
- 1 mm gasket tape under the flanges (tweeter: cut to 3 mm width)
- Butyl mat on the inside walls

Tighten crosswise,
hand-tight only!

7 Damping

Damping – side section (front on the left)



■ Butyl damping mat: side walls, top, bottom, back panel (not on the PR ring or the PCB)

■ Loose damping wool, approx. 30 g: upper third + bottom between woofer and PCB

Do not stuff! Keep approx. 2 cm free around the passive radiator and the PCB.

Loose damping wool (approx. 30 g):
- upper third
- bottom between woofer and PCB

Do not stuff!
Keep 2 cm free around the radiator and the PCB.

Stick the antenna inside, at the top of a side wall.

8 Close the back panel



6× M3×10

Before:

- Wires to terminals L / R / PWR
- 2 mm gasket tape all around the body edge, pierce it at the pillars

Tighten crosswise.

Test: press the woofer gently → the rear radiator moves out.

9 Fit the front grille – done







4× M3×12 grille

Screw heads sit recessed in the grille · rubber feet underneath
Keep at least 3 cm from the wall (radiator at the back)